INKLOOM AI HACKATHON • CODENEXUS INDIA 2026 • ROUND 2: THE BUILD

Brand Rule Lab — Master Product Guide

Comprehensive feature walkthrough, UI manual & architectural specification

Brand Rule Lab is an intelligent pre-publication brand governance platform designed for consumer-facing brands. It pairs deterministic Python rules with NVIDIA NIM claim extraction and fact ranking, Google Gemini fallback, and contextual tone analysis, backed by an immutable fact-version ledger and an append-only audit trail.

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Focus Domain: India D2C Personal Care & Beauty

Verified Status: 216 Tests Passed (100% Green) • Live NVIDIA NIM & Gemini AI

1 DRAFT → 2 CHECK → 3 SUGGEST → 4 APPROVE → 5 LEDGER → 6 MONITOR

The Real Problem (ASCI & Peer Data)

REGULATORY RISK

In high-growth consumer marketing, brands publish rapidly across agencies and generative AI tools. Unsubstantiated claims and off-brand visuals lead to severe scrutiny:

90%	82%	1–4 d
ASCI CASES NEEDED EDITS	SCRUTINISED IN DIGITAL ADS	PEER APPROVAL DELAY

> *"Annoying part design ka approval ni milta... senior deny krdete late factual edit pe."* — Primary peer research.

Subjective feedback ("make it pop"), late factual edits, and unrecorded verbal preferences cause massive rework cycles.

Our Innovation: The Claim-to-Fact Ledger

CORE WEDGE

Conventional tools offer subjective chatbot reviews without evidence lineage. **Brand Rule Lab** ties every approved claim to a verifiable evidence record:

- **Stable Identifier (`stable_claim_id`):** A deterministic 12-character hex ID minted upon approval.
- **Normalized SHA-256 Hash:** Immutable fingerprint of the exact approved wording.
- **Fact-Version Lineage (`fact_version_id`):** Bound to the exact approved fact version (e.g. Clinical Study F-112 v3).
- **Deterministic Drift Cascade:** When evidence changes or expires, dependent claims are flagged in **1 SQL update with 0 LLM calls**.

> **Strict Operational Boundary:** `UNLINKED`, `NOT_OBSERVED`, and `REVIEW_REQUIRED` remain separate states. A missing match is never quietly treated as compliant.

Layered System Architecture

ZERO SINGLE POINT OF FAILURE

1. Presentation Layer

Streamlit 1.66.1 multi-role dashboard. Real-time sidebar role switcher (Creator, Reviewer, Fact Owner). Interactive highlighted text spans, side-by-side diff editor, and live governance metrics.

2. Hybrid Enforcement Layer

Python Rules: Deterministic evaluation of terms, regex, dates, ranges, contrast.
AI Engine: NVIDIA NIM primary (Nemotron-3-Nano) with Gemini fallback for claim extraction and semantic ranking.

3. Persistence & Ledger Layer

SQLite 3 in WAL mode with strict foreign keys. Immutable `fact_versions`, append-only `approved_claims` snapshots, and tamper-evident `review_events` audit logs.

The Complete UI Walkthrough: What Every Page Does & How to Use It

A detailed breakdown of all 10 views accessible via the sidebar role switcher

Page Name	Allowed Roles	What This Page Does	How to Use It Step-by-Step
1. Overview	ALL ROLES	First-use welcome hub, setup progress checklist, database health status, and quick-start scenario links.	Select your demo role from the sidebar. Check the system readiness checklist and click into synthetic demo scenarios.
2. Brand Knowledge	CREATOR / REVIEWER	Onboards raw brand assets (PDF, DOCX, XLSX, TXT, images). Performs local text extraction and OCR. Extracts source-quoted facts & guidelines with explicit AI consent.	Upload a brand manual or clinical study. Enable AI consent checkbox. Click "*Extract Proposals*". Inspect extracted source quotes and page numbers.
3. Guidelines & Rule Readiness	REVIEWER ONLY	Audits guideline proposals into 4 readiness classes (`CODE_CHECKABLE`, `AI_ASSISTED`, `HUMAN_JUDGEMENT`, `UNCLEAR`). Compiles predicates and runs automated self-tests.	Review pending proposals. Assign readiness class. For `CODE_CHECKABLE`, pick predicate (`term_list`, etc.), run positive/negative tests, and click "*Approve Rule*".
4. Facts & Claims	FACT OWNER / REVIEWER	Central registry of approved product facts. Shows statement, evidence reference, valid date window, market scope, and version history.	Filter by product. Review fact versions. Click "*New Fact Version*" to update clinical evidence, or expire outdated facts to trigger drift.
5. Assets	CREATOR / FACT OWNER	Registers and manages approved product images, logos, and packaging renders. Computes exact SHA-256 and pHash fingerprints.	Upload approved image assets. View SHA-256 identity hashes. Set approval status so drafts can be verified against canonical creative.
6. New Review	CREATOR ONLY	Draft intake console. Select product, target market (e.g. IN), channel (e.g. instagram), and optional campaign brief. Accepts pasted copy or document upload.	Select "Monsoon Clear Skin Face Wash". Paste marketing copy or upload PDF/image. Toggle AI consent. Click "*Run Brand & Claim Review*".
7. Review Results	CREATOR / REVIEWER	Displays cited findings (Tone V-04, Terminology T-12, Claim F-112, Contrast, Date). Highlights exact text spans. Side-by-side diff editor with one-click corrections.	Inspect red/amber violation tags. Click "*Apply Suggestion*" to automatically replace off-brand phrasing. Save new draft version and submit for approval.
8. Approval Queue	REVIEWER ONLY	Reviewer decision inbox. Lists submitted drafts awaiting sign-off. Shows blocking rule violations, unlinked claims, and reviewer action buttons.	Open submitted draft. Review findings and fact linkages. Click "*Approve*" (mints stable claim ID), "*Request Changes*", or "*Escalate*".
9. Claim Ledger	FACT OWNER / REVIEWER	Immutable cryptographic ledger of all approved claims. Displays `stable_claim_id`, normalized text hash, linked `fact_version_id`, and real-time drift state.	Inspect registered claims. Filter by status: `CURRENT`, `REVIEW_REQUIRED`, or `EXPIRED`. View full audit event history and reviewer signatures.
10. Insights & Analytics	ALL ROLES	Live governance analytics computed directly from the database. Shows rule failure frequencies, turnaround times, and audit activity. Zero fabricated stats.	Analyze recurring brand mistakes (e.g., T-12 BPO infractions) over time to identify areas where brand guidelines or creator training need updating.

Engine Architecture: How Every Check Operates Under the Hood

Deep dive into deterministic predicates, claim extraction, evidence ranking, and tone evaluation

1. The Deterministic Rule Engine (Pure Python)

100% REPRODUCIBLE

Pure Python predicates execute without calling external AI models. They provide microsecond-level pass/fail evaluation, guarantee zero hallucinations, and cannot be bypassed by prompt injection:

Predicate Type	Parameter Configuration	Evaluation Mechanism	Production Use Case
`term_list`	`banned_terms`, `allowed_terms`, `case_sensitive`	Regex boundary search against prohibited words or acronyms.	Disallow "BPO" in consumer copy; require "Benzoyl Peroxide".
`regex`	`pattern`, `must_match` (True/False)	Standardized regex evaluation with bounded execution timeouts.	Enforce mandatory registration numbers or disclaimer formatting.
`numeric_range`	`field_name`, `min_val`, `max_val`	Extracts numeric percentages or dosages and validates bounds.	Validate active ingredient concentrations (e.g. Salicylic Acid ≤ 2.0%).
`date_window`	`valid_from`, `expires_at`	Compares published or scheduled dates against approved windows.	Verify festive promo offers do not outlive approved campaign dates.
`required_text`	`substrings`, `all_required`	Verifies exact presence of mandatory legal / statutory warnings.	Mandate "Patch test recommended" on exfoliating skincare acids.
`wcag_contrast`	`fg_color`, `bg_color`, `min_ratio`	Calculates relative luminance and contrast ratio per WCAG 2.1.	Flag text overlays on social graphics with contrast < 4.5:1.
`asset_hash`	`expected_sha256`	Computes cryptographic SHA-256 hash of submitted creative file.	Verify social ad visual matches the exact approved product packshot.

2. AI Claim Extraction Pipeline

OFFSET PRESERVED

To prevent LLM hallucination of quotes or positions, extraction is strictly bounded:

1. **Source Segmentation:** Python segments the text into sentence and clause candidates with exact character start/end offsets.
2. **Candidate Selection:** External AI receives candidate IDs (e.g. `C0001`) and labels the claim type hint.
3. **Deterministic Slicing:** Python slices the draft using verified offsets. The model is physically incapable of inventing non-existent spans.

3. Evidence Ranking & Tone Assessment

SCHEMA VALIDATED

Once claims are extracted, evidence ranking matches approved facts:

- **SQL Scoped Filter:** SQLite filters approved facts strictly by product ID and market. Claims never see facts from other brands.
- **Four Evidence States:** AI ranks relationship into `SUPPORTS`, `CONTRADICTS`, `RELATED_NOT_EVIDENCE`, or `NO_MATCH`.
- **Tone & Brief:** Advisory tone analysis inspects brand voice rules (V-04) and campaign briefs. AI suggestions require human approval.

Governance Lifecycle: Rule Readiness & The Life of a Claim

How human reviewers gate automated rules and how claims travel from draft to cryptographic ledger

The Four Rule Readiness Classes

AUDIT GATE

Extracted brand rules are never automatically enforced. A human reviewer audits each proposal against its source document quotation and assigns one of four strict classifications:

1. CODE_CHECKABLE (Deterministic Active Rule)

Rule can be cleanly converted into a whitelisted Python predicate. Reviewer specifies parameters (e.g. banned term "BPO") and must run automated positive and negative self-tests before activation.

2. AI_ASSISTED (Advisory Contextual Guidance)

Nuanced guidelines such as brand voice, storytelling style, and emotional resonance. The rule provides advisory suggestions during review but is never compiled into a blocking rule.

3. HUMAN_JUDGEMENT (Manual Decision Gate)

Aesthetic balance, cultural appropriateness, or subjective brand fit. The system bypasses automated evaluation and routes the content directly to the reviewer queue.

4. UNCLEAR (Clarification Required)

Contradictory or poorly phrased guidelines. Flagged as needing clarification; strictly blocked from enforcement until the brand owner documents a written resolution.

The Life of an Approved Claim

LEDGER TRACEABILITY

1. **Draft Submission:** Creator drafts copy and submits for review.
2. **Pre-Publication Check:** Python predicates and AI extraction evaluate the copy.
3. **Human Approval:** Reviewer inspects findings and signs off. AI cannot approve.
4. **Minting Stable ID:** System generates a 12-char hex `stable_claim_id` and SHA-256 hash.
5. **Snapshot Commitment:** Stores approved wording, product ID, market, reviewer name, timestamp, and linked `fact_version_id`.

Drift Propagation & Expiry Sweeps

ZERO-LLM MAINTENANCE

When clinical facts evolve, the claim ledger updates automatically:

- **Fact Version Bump:** If Fact Owner updates Fact F-112 to version 4, all claims linked to versions < 4 are flagged as `REVIEW_REQUIRED`.
- **Fact Expiration:** When a fact exceeds its `expires_at` date, an atomic sweep transitions dependent claims to `EXPIRED`.
- **Targeted Isolation:** Unrelated claims remain `CURRENT`. Zero web crawling or manual re-auditing required.

Demonstration Scenarios, Deployment & Future Roadmap

Concrete execution walkthrough on Monsoon Beauty Co. and next-phase additions

Live Demonstration Scenarios (Monsoon Beauty Co.)

PRE-LOADED FIXTURES

Domain / Rule	Offending Draft Copy	System Finding & Enforcement	Preserved Intent Suggestion
Tone (V-04)	"*Say goodbye to acne forever!*"	FAIL Absolute promise conflicts with approved calm/credible voice rule V-04.	Suggest qualified phrasing: "*Helps clear blemishes*"; reviewer confirms.
Terminology (T-12)	"*Our 2.5% BPO face wash*"	FAIL Predicate T-12 disallows "BPO" in consumer-facing copy.	One-click intent replacement with approved INCI term: "*Benzoyl Peroxide*".
Claim vs. Fact (F-112)	"*Clears breakouts in 3 days*"	UNLINKED Approved clinical trial MS-2024-F112 observes 4-week timeline, not 3 days.	Suggest: "*Within 4 weeks of twice-daily use*" citing Clinical Study MS-2024-F112.
Campaign Dates	"*Diwali sale ends 15 Nov*"	FAIL `date_window` predicate check fails; approved brief campaign expires 12 Nov.	Flag for campaign owner: renew approved fact or update copy date.
Visual Contrast	Banner graphic contrast ratio 2.5:1	FAIL Fails WCAG AA 4.5:1 threshold for normal text legibility.	Adjust background or font color to achieve contrast ≥ 4.5:1.

Technical Stack & Deployment

DOCKER & CLOUD READY

- **UI:** Streamlit 1.66.1 (multi-role views, reactive diff & audit inspectors).
- **Primary AI:** NVIDIA NIM HTTPS API (`nemotron-3-nano-omni-30b-a3b-reasoning`).
- **Fallback AI:** Google Gemini API (`gemini-3.8-flash`) with auto-switchover.
- **Local Parsing:** PyMuPDF (PDF), python-docx (DOCX), openpyxl (XLSX), Tesseract OCR (images).
- **Testing & Quality:** 216 pytest unit/integration tests (**216 passed, 0 failed**).
- **Deployment:** Multi-stage Dockerfile with Debian Tesseract OCR and automated database seeding on boot.

Three Major Round 3 Improvements

FUTURE SCOPE

1. Pre-Flight CMS & Social Gatekeeper: Direct bi-directional webhooks for Shopify, WordPress, and Hootsuite that intercept publication if a valid `stable_claim_id` is missing or drifted.

2. Multimodal Packaging & Video Frame Inspection: Temporal analysis using Whisper for audio disclaimers and frame extraction for video overlays and 3D packaging renders.

3. Enterprise RBAC & Cryptographic Merkle Export: Single Sign-On (OIDC/SAML) and mathematically tamper-evident Merkle-tree claim exports for ASCI/FTC legal compliance audits.